

RAPHAEL HUANG – CURRICULUM VITAE

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RESEARCH INTERESTS

VLSI, place and route algorithms; Formal verification of hardware systems, DFT; FPGA design, HDLs, heterogenous computing systems; Connected and automated vehicles, modular robotics.

EDUCATION

University of California, Santa Cruz Master of Science in Computer Science and Engineering GPA: 4.00	September 2025 - Present
University of California, Santa Cruz Bachelor of Science, <i>magna cum laude</i> in Computer Engineering GPA: 3.93	September 2021 - June 2025

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant , UC Santa Cruz, Baskin Engineering - Coordinated logistical and technical aspects of a ~150-student Introduction to Logic Design course. - Piloted the course's transition to a new open-source hardware toolchain, improving equitability. - Oversaw the creation of homework and exams, ensuring consistent evaluation of student learning. - Delivered occasional lectures and provided one-on-one assistance to students in lab sections, helping to debug large structural Verilog programs in Vivado, and clarify complex concepts.	September 2025 - Present
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Connected and Automated Vehicle R&D Intern , Argonne National Laboratory - Transitioned LARS (Longitudinal Automated Replay System), a real-time vehicle energy analytics tool, from Simulink to a modular ROS 2 stack, enabling rapid cross-vehicle deployment. - Validated the system through simulation and on-road testing on a Cadillac LYRIQ, implementing an Iterative Learning Control algorithm that reduced root-mean-square tracking error by ~65%. - Delivered an accessible platform to support research and workforce development for the collegiate EcoCAR EV Challenge and the broader CAV community [R2].	June 2025 - August 2025
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Instructional Design Research Assistant , UC Santa Cruz, Hardware Systems Collective - Redesigned the curriculum for CSE 100, Introduction to Logic Design, focusing on accessible, equitable, and pedagogically-sound coursework. - Created homeworks and labs that enabled remote completion of coursework that enabled course enrollment to scale to over 140 students per quarter (previously ~80).	Jan - July 2025
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Lead Instructor , iDTech Camps, Stanford University - Led 11 instructors teaching AI/ML in Python and VEX Robotics to 100+ students while adapting curriculum for a pilot program for international learners from China and Latin America.	July - August 2024
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RESEARCH AND PROJECTS

Formal Methods for VLSI Timing Optimization - Developing a formal methods approach to correct Static Timing Analysis (STA) violations by integrating a Z3 OMT solver with the OpenROAD open-source ASIC design flow. - Created Python and Tcl scripts to parse timing reports, generate logical constraints for optimal cell resizing, and programmatically modify netlists [R1].	September 2025 - Present
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Equitable FOSS Logic Design Curriculum Development

January 2025 - Present

- Designing a study to quantify the impact of shifting from proprietary to open-source tools (e.g. Yosys, Verilator) on student learning outcomes, scalability, accessibility, and equity.

Vivaldi: A SystemVerilog FPGA Audio Synthesizer

January - March 2025

- Wrote large-scale RTL in **SystemVerilog** for a Artix-7 FPGA to implement a multi-oscillator audio synthesizer; automated the verification and synthesis flow and validated the design with a comprehensive **DPI-C** testbench.

Digital Circuit Design Animations and SystemVerilog Youtube Tutorials

September 2024 - March 2025

- Collaborated with Ethan Sifferman (UC Santa Cruz) to work towards YouTube tutorials on Verilog/SystemVerilog topics. Drafted scripts, and created `manim-digital`, an open-source combinational and sequential digital logic animation library.

RISC-V on PYNQ

September 2024 - March 2025

- Worked with Prof. Dustin Richmond on updating a tutorial that uses web-based Python Jupyter notebooks to teach the user how to create, package, and use a **RISC-V** processor overlay for the Xilinx PYNQ-Z1 board.

FPGA UART ALU

January 2025

- Developed a custom variable-length packet parsing state machine to drive a 32-bit ALU via UART, enabling real-time verification with Python.

RELEVANT COURSEWORK

Graduate (M.S. in CSE)

- Formal Methods (CSE 216)
- Advanced VLSI Design (CSE 222A)
- Computer Architecture (CSE 220)
- Analysis of Algorithms (CSE 201)

Undergraduate (B.S. in CE)

- Logic Design with Verilog (CSE 125)
- Embedded System Design (CSE 121)
- Probability & Statistics (CSE 107)
- Signals & Systems (ECE 103)

HONORS, AWARDS, AND FELLOWSHIPS

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| [A4] | 2026-2027 | Graduate Pedagogy Fellowship, Teaching and Learning Center, UC Santa Cruz |
| [A3] | 2025 | Impact Argonne Award: Energy Secretary Visit, Argonne National Lab |
| [A2] | 2021-2025 | Dean's Honors, UC Santa Cruz |
| [A1] | 2021 | UC Regents Scholar, Regents of the University of California |

TECHNICAL REPORTS

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| [R1] | C. Baker, R. Huang, W. Ali, A. Muddamsetti, <i>Optimal Cell Resizing in OpenROAD-flow-scripts Using Optimization Modulo Theories</i> , Baskin School of Engineering, University of California, Santa Cruz, Santa Cruz, California, 2025. |
| [R2] | Goberville, Nick and Huang, Raphael, <i>Modernizing LARS: Open-source, Modular, and Programmatically Flexible Scenario Replay</i> , Argonne National Lab, Science Undergraduate Laboratory Internships (SULI), Lemont, Illinois, 2025. |
| [R3] | R. Huang, D. Lee, C. Baker, V. Harkins, and G. Austin, <i>PillPal: A Smart Medication Dispensing System</i> , Baskin School of Engineering, University of California, Santa Cruz, Santa Cruz, CA, 2025. |